In [ ]:

```
# Model name: microsoft/resnet-18
# Task: Image classification
# Platform: Hugging Face
# Library: Transformers
#Execution: Local Jupyter Notebook
```

In [15]:

```
import importlib.util
import sys

print("Python executable:", sys.executable)

# Check required libraries
for package in ["torch", "transformers", "PIL"]:
    installed = importlib.util.find_spec(package) is not None
    print(f"{package}: {'Installed' if installed else 'Missing'}")
```

```
Python executable: C:\ProgramData\anaconda3\python.exe
torch: Installed
transformers: Installed
PIL: Installed
```

In [16]:

```
# Install Hugging Face Transformers
%pip install transformers
```

```
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: transformers in C:\Users\Gayathri R\AppData\Roaming\Pytho
n\Python314\site-packages (5.20.2)
Requirement already satisfied: huggingface-hub<3.0,>=1.31.0 in C:\Users\Gayathri R\AppDa
ta\Roaming\Python\Python314\site-packages (from transformers) (2.2.0)
Requirement already satisfied: numpy>=1.17 in C:\Users\Gayathri R\AppData\Roaming\Python
\Python314\site-packages (from transformers) (2.5.4)
Requirement already satisfied: packaging>=20.0 in C:\ProgramData\anaconda3\Lib\site-pack
ages (from transformers) (26.0)
Requirement already satisfied: pyyaml>=5.1 in C:\ProgramData\anaconda3\Lib\site-packages
(from transformers) (6.0.3)
Requirement already satisfied: regex>=2025.10.22 in C:\ProgramData\anaconda3\Lib\site-pa
ckages (from transformers) (2026.5.9)
Requirement already satisfied: tokenizers<0.24.0,>=0.23.1 in C:\Users\Gayathri R\AppData
\Roaming\Python\Python314\site-packages (from transformers) (0.23.3)
Requirement already satisfied: typer in C:\ProgramData\anaconda3\Lib\site-packages (from
transformers) (0.25.1)
Requirement already satisfied: safetensors>=0.8.0 in C:\Users\Gayathri R\AppData\Roaming
\Python\Python314\site-packages (from transformers) (0.8.0)
Requirement already satisfied: tqdm>=4.60 in C:\ProgramData\anaconda3\Lib\site-packages
(from transformers) (4.68.2)
Requirement already satisfied: click<9.0.0,>=8.4.2 in C:\Users\Gayathri R\AppData\Roamin
g\Python\Python314\site-packages (from huggingface-hub<3.0,>=1.31.0->transformers) (8.5.
0)
Requirement already satisfied: filelock>=3.10.0 in C:\ProgramData\anaconda3\Lib\site-pac
kages (from huggingface-hub<3.0,>=1.31.0->transformers) (3.29.4)
Requirement already satisfied: fsspec>=2023.5.0 in C:\ProgramData\anaconda3\Lib\site-pac
kages (from huggingface-hub<3.0,>=1.31.0->transformers) (2026.4.0)
Requirement already satisfied: hf-xet<2.0.0,>=1.6.0 in C:\Users\Gayathri R\AppData\Roami
ng\Python\Python314\site-packages (from huggingface-hub<3.0,>=1.31.0->transformers) (1.
7.0)
Requirement already satisfied: httpx2<3,>=2.0.0 in C:\Users\Gayathri R\AppData\Roaming\P
```

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```
ython\Python314\site-packages (from huggingface-hub<3.0,>=1.31.0->transformers) (2.15.1)
Requirement already satisfied: typing-extensions>=4.1.0 in C:\ProgramData\anaconda3\Lib
\site-packages (from huggingface-hub<3.0,>=1.31.0->transformers) (4.16.0)
Requirement already satisfied: anyio>=4.10 in C:\ProgramData\anaconda3\Lib\site-packages
(from httpx2<3,>=2.0.0->huggingface-hub<3.0,>=1.31.0->transformers) (4.12.1)
Requirement already satisfied: httpcore2==2.15.1 in C:\Users\Gayathri R\AppData\Roaming
\Python\Python314\site-packages (from httpx2<3,>=2.0.0->huggingface-hub<3.0,>=1.31.0->tr
ansformers) (2.15.1)
Requirement already satisfied: idna>=3.18 in C:\ProgramData\anaconda3\Lib\site-packages
(from httpx2<3,>=2.0.0->huggingface-hub<3.0,>=1.31.0->transformers) (3.18)
Requirement already satisfied: truststore>=0.10 in C:\ProgramData\anaconda3\Lib\site-pac
kages (from httpx2<3,>=2.0.0->huggingface-hub<3.0,>=1.31.0->transformers) (0.10.4)
Requirement already satisfied: h11>=0.16 in C:\ProgramData\anaconda3\Lib\site-packages
(from httpcore2==2.15.1->httpx2<3,>=2.0.0->huggingface-hub<3.0,>=1.31.0->transformers)
(0.16.0)
Requirement already satisfied: colorama in C:\ProgramData\anaconda3\Lib\site-packages (f
rom tqdm>=4.60->transformers) (0.4.6)
Requirement already satisfied: shellingham>=1.3.0 in C:\ProgramData\anaconda3\Lib\site-p
ackages (from typer->transformers) (1.5.4)
Requirement already satisfied: rich>=13.8.0 in C:\ProgramData\anaconda3\Lib\site-package
s (from typer->transformers) (15.0.0)
Requirement already satisfied: annotated-doc>=0.0.2 in C:\ProgramData\anaconda3\Lib\site
-packages (from typer->transformers) (0.0.4)
Requirement already satisfied: markdown-it-py>=2.2.0 in C:\ProgramData\anaconda3\Lib\sit
e-packages (from rich>=13.8.0->typer->transformers) (4.0.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in C:\ProgramData\anaconda3\Lib\s
ite-packages (from rich>=13.8.0->typer->transformers) (2.20.0)
Requirement already satisfied: mdurl~=0.1 in C:\ProgramData\anaconda3\Lib\site-packages
(from markdown-it-py>=2.2.0->rich>=13.8.0->typer->transformers) (0.1.2)
Note: you may need to restart the kernel to use updated packages.
```

In [17]:

```
# Install image processing library
%pip install pillow
```

```
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: pillow in C:\Users\Gayathri R\AppData\Roaming\Python\Pyth
on314\site-packages (13.0.0)
Note: you may need to restart the kernel to use updated packages.
```

In [18]:

```
# Install deep learning library
%pip install torch
```

```
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: torch in C:\Users\Gayathri R\AppData\Roaming\Python\Pytho
n314\site-packages (2.14.1)
Requirement already satisfied: filelock in C:\ProgramData\anaconda3\Lib\site-packages (f
rom torch) (3.29.4)
Requirement already satisfied: typing-extensions>=4.10.0 in C:\ProgramData\anaconda3\Lib
\site-packages (from torch) (4.16.0)
Requirement already satisfied: setuptools>=77.0.3 in C:\ProgramData\anaconda3\Lib\site-p
ackages (from torch) (83.0.0)
Requirement already satisfied: sympy>=1.13.3 in C:\ProgramData\anaconda3\Lib\site-packag
es (from torch) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in C:\ProgramData\anaconda3\Lib\site-pack
ages (from torch) (3.6.1)
Requirement already satisfied: jinja2 in C:\ProgramData\anaconda3\Lib\site-packages (fro
m torch) (3.1.6)
Requirement already satisfied: fsspec>=0.8.5 in C:\ProgramData\anaconda3\Lib\site-packag
```

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```
es (from torch) (2026.4.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in C:\ProgramData\anaconda3\Lib\site-p
ackages (from sympy>=1.13.3->torch) (1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in C:\ProgramData\anaconda3\Lib\site-pack
ages (from jinja2->torch) (3.0.3)
Note: you may need to restart the kernel to use updated packages.
```

In [19]:

```python
# Import and check library versions
import torch
import transformers
from PIL import Image

print("PyTorch:", torch.__version__)
print("Transformers:", transformers.__version__)
print("All libraries are working!")
```

```
PyTorch: 2.14.1+cpu
Transformers: 5.20.2
All libraries are working!
```

In [20]:

```python
# Load ResNet-18 for image classification
from transformers import pipeline

classifier = pipeline(
    "image-classification",
    model="microsoft/resnet-18",
    device=-1
)

print("Model loaded successfully!")
```

```
Could not reach the Hub ([WinError 10054] An existing connection was forcibly closed by
the remote host). Using cached commit hash for 'microsoft/resnet-18'.
Loading weights:   0%|          | 0/122 [00:00<?, ?it/s]
Model loaded successfully!
```

In [22]:

```python
# Check whether the image exists
from IPython.display import display
from PIL import Image
from pathlib import Path

image_path = Path("elephant.jpg")

print("Current folder:", Path.cwd())
print("Image exists:", image_path.exists())
```

```
Current folder: D:\basic python
Image exists: False
```

In [23]:

```python
# Select and display an image from your pc
from tkinter import Tk, filedialog
from PIL import Image
from IPython.display import display

root = Tk()
root.withdraw()
root.attributes("-topmost", True)
```

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```
file_path = filedialog.askopenfilename(
    title="Select an image",
    filetypes=[
        ("Image files", "*.jpg *.jpeg *.png"),
        ("All files", "*.*")
    ]
)

root.destroy()

if file_path:
    image = Image.open(file_path).convert("RGB")
    display(image)
    print("Image loaded successfully!")
else:
    print("No image selected.")
```



```
Image loaded successfully!
```

In [24]:

```
# Predict the top 3 image categories
results = classifier(image, top_k=3)

for result in results:
    print(f"{result['label']}: {result['score']:.2%}")
```

```
tusker: 55.37%
African elephant, Loxodonta africana: 37.73%
Indian elephant, Elephas maximus: 6.90%
```

In [ ]:

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```
```

In [ ]:

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```

In [ ]:

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```

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